Sri Sivasubramaniya Nadar College of Engineering, Chennai

(An autonomous Institution affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering	**Semester**	VI
Subject Code & Name	UCS2612– Machine Learning Laboratory		
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Experiment 7: Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Clustering

Objective

To implement and analyze the performance of clustering algorithms on the Human Activity Recognition dataset:

- **Model A:** K-Means Clustering.
- **Model B:** DBSCAN (Density-Based Spatial Clustering of Applications with Noise).
- **Model C:** Hierarchical Agglomerative Clustering (HAC).

Students must visualize and compare the clusters formed by each algorithm, report results, and analyze performance.

Dataset

Download: Human Activity Recognition Using Smartphones Dataset

- 30 volunteers (aged 19–48). - 6 activities: WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, LAYING. - Sensors: 3-axis accelerometer and gyroscope (50 Hz). - Preprocessed into windows of 2.56s (128 readings/window) with feature extraction from time and frequency domain.

Theory

K-Means Clustering

- Objective: Partition data into k clusters by minimizing within-cluster variance. - Distance metric: Euclidean. - Algorithm:

1. Initialize k centroids.
2. Assign points to nearest centroid.
3. Update centroids as mean of assigned points.
4. Repeat until convergence.

Elbow Method for Choosing k

- Compute **Within-Cluster Sum of Squares (WCSS)** for different k values. - Plot k vs. WCSS.
- The "elbow point" (where reduction in WCSS slows down) suggests the optimal k.

Table 1: K-Means Elbow Method Results

Number of Clusters (k)	**WCSS (Inertia)**	**Silhouette Score**
2	----	----
3	----	----
4	----	----
5	----	----
6	----	----
7	----	----
8	----	----

Students must:

- Fill in the table above after running experiments.
- Plot k vs. WCSS (Elbow curve).
- Plot k vs. Silhouette Score.
- Select and justify the best k.

DBSCAN

- Density-based clustering algorithm. - Parameters: ϵ (neighborhood radius), $minPts$ (minimum points). - Core points, border points, and noise are identified. - Advantage: Can discover clusters of arbitrary shape and handle noise.

Hierarchical Clustering

- Builds hierarchy of clusters using either:

- Agglomerative (bottom-up merging).
- Divisive (top-down splitting).

- Linkage criteria: Single, Complete, Average, Ward's method. - Represented by a dendrogram.

Steps for Implementation

1. Load the dataset and preprocess:
    - Handle missing values.
    - Encode categorical labels.
    - Apply normalization/standardization.

2. Perform exploratory data analysis (EDA):
   - Visualize feature distributions.
   - Apply dimensionality reduction (PCA/t-SNE) for visualization.
3. Apply clustering algorithms:
   - **K-Means:** Use Elbow method to choose k.
   - **DBSCAN:** Tune ϵ and $minPts$.
   - **Hierarchical:** Apply Agglomerative clustering with Ward's linkage.
4. Visualize clusters in 2D using PCA/t-SNE.
5. Compare results with ground truth activity labels (although clustering is unsupervised).
6. Analyze performance using clustering metrics.

Evaluation Metrics

- **Internal:** Silhouette Score, Davies–Bouldin Index, Calinski–Harabasz Index.
- **External (w.r.t. true labels):** Adjusted Rand Index (ARI), Normalized Mutual Information (NMI).
- Confusion matrix (optional, by mapping clusters to activities).

Graphs and Visualization

Students should produce:

- Elbow curve: k vs. WCSS.
- Silhouette curve: k vs. Silhouette Score.
- Scatter plots of clusters (after PCA/t-SNE).
- Dendrograms for hierarchical clustering.
- Bar plots of evaluation metrics across algorithms.

Observation Questions

- Which algorithm produced the most meaningful clusters? Why?
- How sensitive was K-Means to the choice of k?
- Did DBSCAN detect noise or small clusters effectively?
- How does linkage choice (single/complete/ward) affect hierarchical clustering?
- Which internal metric best matched your visual intuition of cluster quality?

Report Checklist

- Aim and Objective
- Preprocessing Steps
- K-Means with Elbow Method Results
- Clustering Implementation (K-Means, DBSCAN, Hierarchical)
- Evaluation Metrics and Results
- Graphs and Visualizations
- Observations and Comparative Analysis